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```
# Import required libraries
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.datasets import mnist

# Load MNIST dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Print dataset shapes
print("Training data shape:", X_train.shape)
print("Training labels shape:", y_train.shape)
print("Testing data shape:", X_test.shape)
print("Testing labels shape:", y_test.shape)

# Display first 5 images
plt.figure(figsize=(10, 4))
for i in range(5):
    plt.subplot(1, 5, i + 1)
    plt.imshow(X_train[i], cmap='gray')
    plt.title(f"Label: {y_train[i]}")
    plt.axis('off')
plt.show()

# Flatten images
X_train_flat = X_train.reshape(X_train.shape[0], 28 * 28)
X_test_flat = X_test.reshape(X_test.shape[0], 28 * 28)

# Normalize pixel values
X_train_flat = X_train_flat / 255.0
X_test_flat = X_test_flat / 255.0

print("Flattened training data shape:", X_train_flat.shape)
print("Flattened testing data shape:", X_test_flat.shape)

from sklearn.model_selection import train_test_split

X_train_split, X_val, y_train_split, y_val = train_test_split(
    X_train_flat, y_train, test_size=0.2, random_state=42
)

from sklearn.tree import DecisionTreeClassifier

# Initialize the Decision Tree classifier
dt_classifier = DecisionTreeClassifier(
    criterion='gini',
    max_depth=20,
    random_state=42
)

# Train the model
dt_classifier.fit(X_train_flat, y_train)

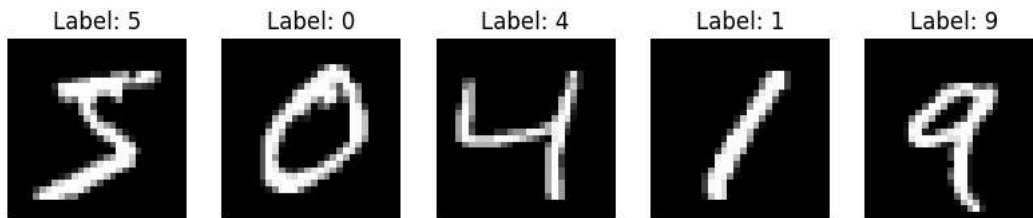
from sklearn.metrics import accuracy_score

# Predict on test data
y_pred = dt_classifier.predict(X_test_flat)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Decision Tree Model Accuracy:", accuracy)
```

```
Training data shape: (60000, 28, 28)
Training labels shape: (60000,)
Testing data shape: (10000, 28, 28)
Testing labels shape: (10000,)
```



```
Flattened training data shape: (60000, 784)
Flattened testing data shape: (10000, 784)
Decision_Tree_Model Accuracy: 0.8818
```

Code Summary

This program implements a Decision Tree classifier to classify handwritten digits from the MNIST dataset.

*Importing Libraries

Required libraries such as numpy, matplotlib, tensorflow.keras.datasets, and sklearn are imported for data handling, visualization, model training, and evaluation.

*Loading the Dataset

The MNIST dataset is loaded using `mnist.load_data()`, which provides:

*Training images and labels

Testing images and labels Each image is of size 28×28 pixels, and labels range from 0 to 9.

*Data Visualization

A few sample images are displayed using matplotlib to understand the handwritten digit patterns present in the dataset.

*Data Preprocessing

Each 28×28 image is flattened into a one-dimensional array of 784 features so that it can be used by the Decision Tree algorithm.

Pixel values are normalized by dividing by 255 to bring them into the range 0 to 1, improving training efficiency.

*Model Creation and Training

A `DecisionTreeClassifier` is initialized using the Gini index as the splitting criterion. The model is trained using the flattened and normalized training data.

*Prediction and Evaluation

The trained model predicts digit labels for the test dataset. The accuracy score is calculated to evaluate the performance of the classifier.

*Result

The Decision Tree model achieves reasonable accuracy in classifying handwritten digits, demonstrating successful application of machine learning on image data.